

Advanced Set theoretic Concepts.

Cartesian Products & Relations.

$A, B \subseteq U \rightarrow$ Cartesian product / cross product is denoted as $A \times B$.

$$A \times B = \{ \underline{(a, b)} \mid a \in A, b \in B \}.$$

$\searrow$ ordered pair.

Computing. Computing.

$$\begin{array}{ccc} (A \times B) & \rightarrow & B \times A \\ \downarrow & & \downarrow \\ (a, b) & \neq & (b, a) \end{array}$$

explicit ordering.

$$(a, b), (c, d) \in A \times B \Rightarrow (a, b) = (c, d) \text{ iff } a = c \text{ \& } b = d.$$

$|A| = n$ elements $|B| = m$ elements

$$|A \times B| = mn = |A| \cdot |B|$$

$$|B \times A| = mn = |A| \cdot |B|$$

Although in general

$$A \times B \neq B \times A$$

$$|A \times B| = |B \times A| = |A| \cdot |B|.$$

$$A, B \subseteq U$$

$$S = A \times B.$$

~~$$A \times B \subseteq U$$~~

$$P(S) \not\subseteq U$$

$$n \in \mathbb{Z}^+ \quad n \geq 3$$

$$A_1, A_2, \dots, A_n.$$

Cartesian product: $A_1 \times A_2 \times \dots \times A_n.$

$$\{(a_1, a_2, \dots, a_n) \mid a_i \in A_i, 1 \leq i \leq n\}.$$

→ ordered n -tuples.

$n=3 \Rightarrow$ ordered triples.

E.g.

$$\mathcal{U} = \{1, 2, 3, 4, \dots, 7\}.$$

$$A = \{2, 3, 4\}, \quad B = \{4, 5\}.$$

points on the Cartesian plane.

$$A \times B = \{(2, 4), (2, 5), (3, 4), (3, 5), (4, 4), (4, 5)\}.$$

$$B \times A = \{(4, 2), (5, 2), (4, 3), (5, 3), (4, 4), (5, 4)\}.$$

$$B^2 = B \times B$$

$$\{(4, 4), (4, 5), (5, 4), (5, 5)\}.$$

E.g. An experiment $\mathcal{E}$ is conducted as follows:

- A single die is rolled and its outcome is noted
- An unbiased coin is flipped and its outcome is noted.

Q: Determine the sample space $\mathcal{S}$ of $\mathcal{E}$.

Part 1: $\mathcal{E}_1, \mathcal{I}_1 = \{1, 2, 3, 4, 5, 6\}$.

Part 2: $\mathcal{E}_2, \mathcal{I}_2 = \{H, T\}$.

$\therefore$ Sample space of $\mathcal{E}$ can be expressed
as $\mathcal{I}_1 \times \mathcal{I}_2$

Defⁿ:

① For set $A, B \subseteq \mathcal{U}$ any subset of $A \times B$
is called a relation from A to B.

② An subset of $A \times A$ is called a
binary relation on A.

E.g. $A, B \subseteq \mathcal{U}$ $|A| = n, |B| = m$.

① What is the total no. of relations from
A to B.

② What is the total no. of binary relations on A

Note:

$$A \times \underline{\Phi} = \underline{\Phi}$$

why?

Contradiction $A \times \underline{\Phi} \neq \underline{\Phi}$

$$(a, b) \in A \times \underline{\Phi}$$

$$a \in A$$

&

$$\boxed{b \in \underline{\Phi}}$$

impossible

Contradiction arrived.

$$\therefore A \times \underline{\Phi} = \underline{\Phi}$$

Theorem:

$$(a) A \times (B \cap C) = (A \times B) \cap (A \times C) \rightarrow$$

$$(b) A \times (B \cup C) = (A \times B) \cup (A \times C)$$

$$(c) (A \cap B) \times C = (A \times C) \cap (B \times C)$$

$$(d) (A \cup B) \times C = (A \times C) \cup (B \times C)$$

} H.W.

$$(1) |A \times B| = mn.$$

↳ relations $\rightarrow$ subset.

All subsets $\Rightarrow 2^{mn}$.

$$(2) |A \times A| = n^2 \rightarrow \text{binary relations} \Rightarrow \boxed{2^{n^2}}$$

Relation : ρ or R .

E.g. R is a subset of $N \times N$

$$R = \{(m, n) \mid n = 7m\}.$$

$$(1) (0, 0) \in R. \Rightarrow 0R0 \checkmark \text{ or } 0P0.$$

$$(2) \text{ If } (s, t) \in R \text{ then } (s+1, t+7) \in R.$$

Properties of (binary) relations.

A relation ρ on a set A is called reflexive if $\forall x \in A, (x, x) \in \rho$.

$$A = \{1, 2, 3, 4\}.$$

* $\rho_1 = \{(1, 1), (2, 2), (3, 3), (4, 4)\}$ is reflexive.

$\rho_2 = \{(1, 1), (3, 3), (4, 4)\}$ is not reflexive as $(2, 2)$ is absent.

E.g. $A = \{a_1, a_2, \dots, a_n\}$.

ρ is reflexive.

How many such ρ 's are there if $|A| = n$.

Proof: $A \times (B \cap C) = (A \times B) \cap (A \times C)$

$$(a, b) \in A \times (B \cap C)$$

$$\Rightarrow a \in A \text{ and } b \in B \cap C.$$

$$\Rightarrow a \in A, \quad b \in B \text{ and } b \in C.$$

$$\Rightarrow a \in A, \quad b \in B \text{ and } a \in A \text{ and } b \in C.$$

$$\Rightarrow (a, b) \in A \times B \text{ and } (a, b) \in A \times C.$$

$$\Rightarrow (a, b) \in (A \times B) \cap (A \times C).$$

$\hookrightarrow$ similarly the other side:

$$(a, b) \in (A \times B) \cap (A \times C) \dots$$

$$\text{Count of } \rho = \underline{\underline{2}} \cdot \underline{\underline{n^2 - n}}$$

↓
diagonal elements (a_i, a_i) .

Left $\Rightarrow n^2 - n$. (elements left).
 exclude an element. 0 0
 include an element. 1 1
 $\left[\begin{array}{cccc} 1 & 1 & \dots & \dots \\ 2 \times & 2 \times & \dots & \dots \end{array} \right] \times \underline{\underline{1}} \text{ (diagonal part)}$

$\underbrace{\hspace{10em}}_{n^2 - n \text{ times}}$ $n^2 - n$
 No. of reflexive ~~relations~~ relations = 2

A relation ρ is said to be symmetric on A

if $(x, y) \in \rho \Rightarrow (y, x) \in \rho$ for
 all $x, y \in A$.

E.g. $A = \{1, 2, 3\}$.

$$P_1 = \{ (1, 2), (2, 1), (1, 3), (3, 1) \}$$

$\rightarrow$ symmetric.

$$P_2 = \{ (1, 1), (2, 2), (3, 3) \}$$

$\rightarrow$ symmetric & reflexive.

$$P_3 = \{ (1, 1), (2, 2), (3, 3), (2, 3), (3, 2) \} \rightarrow S \subseteq R.$$

$$P_4 = \{ (1, 1), (2, 3), (3, 3) \} \quad \begin{matrix} X S \\ X R. \end{matrix}$$

H.W.

Count the no. of symmetric relations on A , ~~set~~ where $|A| = n$.